

Ω TB[®] Genome Analysis Report

Sample

Sample ID: Sample

Sample receipt date: 21-06-2021

Sample shipping date: 12-07-2021

Client

Client ID: CLI00

Lab Name and Location: ABC

Contact: ABC Mumbai Maharashtra

Test

Test ID: OmegaTB

Registered on: 20-06-2021

Sample collected on: 21-06-2021

Reported on: 10-02-2022

Report summary for Sample

Mycobacterium tuberculosis was **Detected**.

The Ω TB analysis has categorized the sample as **Pre-extensively Drug-Resistant tuberculosis (Pre-XDR-TB)**.

Drug resistance has been identified for **9** drugs.

Co-infections identified are **NA**.

Mixed infections identified are **NA**.

Sample summary

Sample ID: Sample

Client ID: CLI00

Test ID: OmegaTB

Sample Type: Sputum

Temperature on receipt(Celsius): 4

Date: 21-06-2021

Sample QC: Pass

QC Check 260/280: 1.35

QC Concentration: 0.31

Sequenced From: Culture

Genome sequencing summary

Library Type: 150x2 paired end

Library QC: Suboptimal

Library Date: 21-06-2021

Genome Sequencer: Illumina NovaSeq

Run Date: 18-07-2021

Machine ID - Flowcell ID: NB501457-HKGN5AFX3

Genomic analysis summary

Application: OmegaTB v2.0.1

Run Date: 10-02-2022

QC check: Pass

Application run by: HAPL

Clinical Summary

Mycobacterium tuberculosis detected	Detected	Coverage	240.72
Lineage	EAI	Proportion of genome covered	1.00

Drug Suceptibility Profile

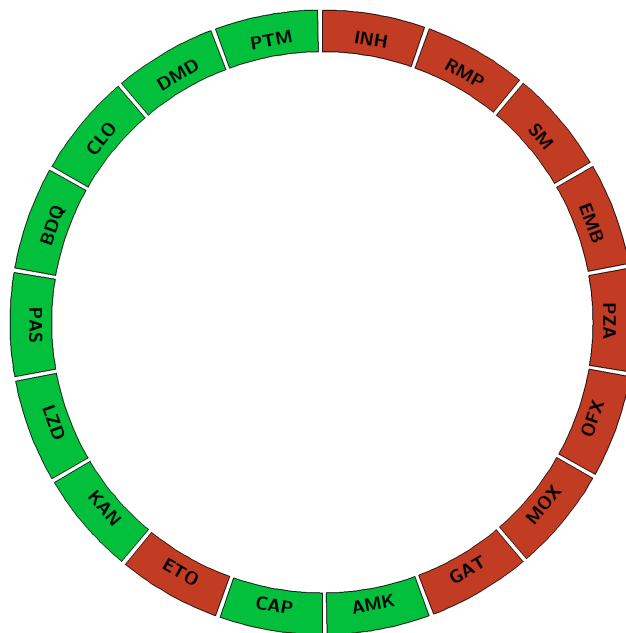
Isoniazid (INH)	Resistant
Rifampicin (RMP)	Resistant
Streptomycin (SM)	Resistant
Ethambutol (EMB)	Resistant
Pyrazinamide (PZA)	Resistant
Ofloxacin (OFX)	Resistant
Moxifloxacin (MOX)	Resistant
Gatifloxacin (GAT)	Resistant
Amikacin (AMK)	No Resistance detected
Capreomycin (CAP)	No Resistance detected
Ethionamide (ETO)	Resistant
Kanamycin (KAN)	No Resistance detected
Linezolid (LZD)	No Resistance detected
Para-aminosalicylic acid (PAS)	No Resistance detected
Bedaquiline (BDQ)	No Resistance detected
Clofazimine (CLO)	No Resistance detected
Delamanid (DMD)	No Resistance detected
Pretomanid (PTM)	No Resistance detected

Clinical Recommendations

Tuberculosis Antibiotic Profile: Pre-extensively Drug-Resistant tuberculosis (Pre-XDR-TB)

Mixed Infection: NA

Antibiogram for Sample



Key:

Green - No Resistance Detected | Red - Resistance Detected

Legends:

INH- Isoniazid, RMP - Rifampicin, SM - Streptomycin, EMB - Ethambutol, PZA - Pyrazinamide, Fluoroquinolones - FQ (OFX - Ofloxacin, MOX - Moxifloxacin, GAT - Gatifloxacin), AMK - Amikacin, CAP - Capreomycin, ETO - Ethionamide, KAN - Kanamycin, LZD -Linezolid, PAS - Para aminosalicylic acid , BDQ - Bedaquiline, CLO - Clofazimine, DMD - Delamanid, PTM - Pretonamid

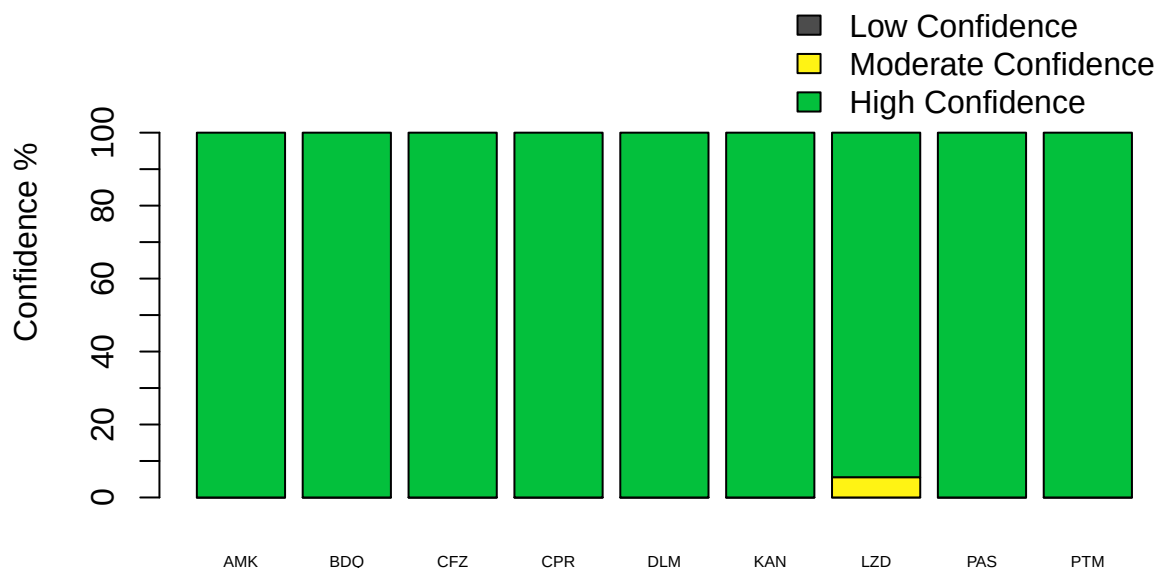
Disclaimer: This is an automated report. The data reported here is based on statistical prediction of variants in the whole genome of the sample processed here. This report is not a phenotypic test of drug resistance. The variants used to predict antibiotic resistance and strain classification are based on genomic variants reported in peer reviewed publications.

Mutations in Sample

Position	Ref Allele	Alternate Allele	Type	Frequency	Coverage	Substitution	Gene Symbol	Gene Name	Drug	PMID	High Confidence SNP
7570	C	T	SNP	69.32	264	Ala90Val (gcg/gTg)	Rv0006	gyrA	FQ	21300839	yes
7581	G	A	SNP	27.86	280	Asp94Asn (gac/Aac)	Rv0006	gyrA	FQ	21562102	yes
761139	C	A	SNP	62.59	139	His445Asn (cac/Aac)	Rv0667	rpoB	RMP	14729930	yes
761155	C	T	SNP	31.85	135	Ser450Leu (tcg/tTg)	Rv0667	rpoB	RMP	21300839	yes
761161	T	C	SNP	70.77	130	Leu452Pro (ctg/cCg)	Rv0667	rpoB	RMP	8913484	yes
781687	A	G	SNP	25.82	244	Lys43Arg (aag/aGg)	Rv0682	rpsL	SM	22646308	yes
1673425	C	T	SNP	99.22	257				INH	21300839	yes
2102240	C	T	SNP	70.38	260	Arg268His (cgc/cAc)	Rv1854c	ndh	INH ETH	11408244	no
2155168	C	G	SNP	31.60	269	Ser315Thr (agc/aCc)	Rv1908c	katG	INH	8878604	yes
2288847	C	G	SNP	31.80	239	Gly132Ala (ggg/gCt)	Rv2043c	pncA	PZA	25336456	yes
4247429	A	G	SNP	70.07	274	Met306Val (atg/Gtg)	Rv3795	embB	EMB	21300839	yes
4248003	A	G	SNP	29.04	272	Gln497Arg (cag/cGg)	Rv3795	embB	EMB	20427375	yes

Disclaimer: This is an automated report. The data reported here is based on statistical prediction of variants in the whole genome of the sample processed here. This report is not a phenotypic test of drug resistance. The variants used to predict antibiotic resistance and strain classification are based on genomic variants reported in peer reviewed publications.

Depth of regions for Sensitive Drugs in Sample



Disclaimer: This is an automated report. The data reported here is based on statistical prediction of variants in the whole genome of the sample processed here. This report is not a phenotypic test of drug resistance. The variants used to predict antibiotic resistance and strain classification are based on genomic variants reported in peer reviewed publications.

ΩTB Drug Resistance Classification Criteria

Multi Drug-Resistant tuberculosis (MDR-TB): Tuberculosis caused by *Mycobacterium tuberculosis* (*M. tuberculosis*) strains that are resistant to at least Isoniazid and Rifampicin.

Pre-extensively Drug-Resistant tuberculosis (Pre-XDR-TB): Tuberculosis caused by *Mycobacterium tuberculosis* (*M. tuberculosis*) strains that fulfil the definition of MDR/Rifampicin Resistant-TB and that are also resistant to any Fluoroquinolones

Extensively Drug-Resistant tuberculosis (XDR-TB): Tuberculosis caused by *Mycobacterium tuberculosis* (*M. tuberculosis*) strains that fulfil the definition of MDR/Rifampicin Resistant-TB and that are also resistant to any Fluoroquinolones and at least one additional Group A drug (The Group A drugs are currently Levofloxacin or Moxifloxacin, Bedaquiline and Linezolid)

Drug resistance detected: Tuberculosis caused by *Mycobacterium tuberculosis* (*M. tuberculosis*) strains that fulfil that is resistant to anti-TB drugs and do not fall in the above mentioned categories

Not Drug Resistance: No Resistance for the anti-TB drugs detected

***** End of the Document *****